

Forecasting USD/INR in the Indian Market with VMD, GDELT Agents, Monte Carlo, and TEPC

COURSE IN CHARGE
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POSTER FORMAT
A4 Portrait

Repository-wide summary of how the project moved from raw news correlation and decomposition toward bounded LLM overlays, market-memory systems, and topology-enabled predictions from chaos.



RAW MARKET SIGNAL

0.010

Goldstein vs IMF3 correlation is nearly zero without structure.

RECOVERED SIGNAL

-0.147

Theme-filtered Tone_Economy regains a meaningful relation to IMF3.

V4 LLM RESULT

64.17%

Gemini V4 direction accuracy on 240 historical hybrid days.

TEPC RESULT

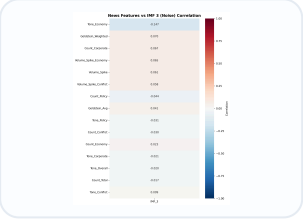
77.5%

Strict TEPC breakout accuracy with macro-F1 of 0.337.

1. PROBLEM

Why the Indian Market Is Hard to Predict

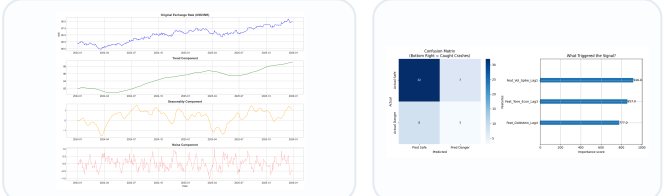
- USD/INR behaves like a managed-float market with RBI intervention, oil shocks, dollar cycles, and delayed news absorption.
- Raw sentiment is weak in isolation: Goldstein vs IMF3 correlation = **0.010**; US average tone vs USD/INR correlation = **0.164**.
- Pattern discovery still finds delayed effects: Goldstein peaks at **9 days** and tone peaks at **12 days**.



2. VMD + STATS

VMD Reframes the Task from Price to Volatility

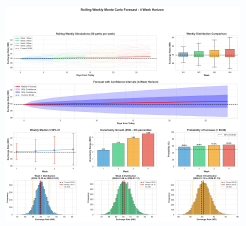
- Three-mode VMD targets **IMF3**, the high-frequency, news-sensitive component of USD/INR.
- Theme buckets used in Phase B: Economy, Conflict, Policy, and Corporate.



Key insight: the market looks random at the raw aggregate level, but specific structure appears once the signal is filtered

3. MONTE CARLO

Monte Carlo Supports a Noise-Dominant Market View



- The branch is strongest as a **distributional risk engine**, not as a single-point predictor.

Recovered evidence: Tone_Economy = -0.147 correlation with IMF3. Gemini V4 classifier reaches 40% precision and 52% recall.

4. GDELT AGENTS

Multi-Agent LLMs Work Only as Bounded Overlays



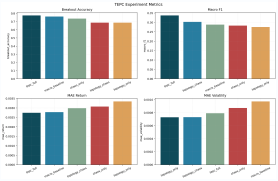
V4 Accuracy Table	Days	MAE	Dir.
V3 hybrid	240	0.2741%	61.25
V4 hybrid	240	0.2695%	64.17
V4 Jan live	21	0.3216%	57.14

Debiasing mattered: symmetric prompting, randomized direction order, adaptive weights, and persona accountability improved V4 over V3.

Regime-switching Monte Carlo gives the most balanced 30-day distribution: median 90.370, std 1.336, 5th to 95th percentile 88.232 to 92.618.

5. TEPC

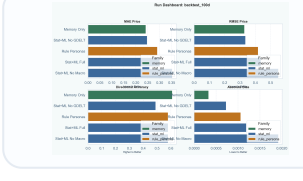
Most Recent Research: Topology-Enabled Predictions from Chaos



- TEPC models USD/INR as one node in a macro-geopolitical network with DXY, BRENT, GOLD, rates, sentiment, Goldstein spread, and geopolitical pressure.
- Rolling correlations create a graph; Laplacian summaries plus coupled Lorenz chaos features become predictive signals.
- tepc_full** wins the strict raw-GDELT run with **77.5% breakout accuracy**, macro-F1 of 0.337, and **0.002746 MAE** return on a bigger black-box model. The strongest path in this repo is a layered pipeline: **VMD + thematic GDELT filtering + probabilistic sanity checks + bounded multi-agent reasoning + TEPC topology/chaos features**.

6. HYBRID TAKEAWAY

TEPC + LLMs for the Win Means Structured State + Controlled Reasoning



TEPC / LLM Accuracy Table	Window	Accuracy
Memory-only state	100d	61.0% dir.
Gemini 2.5 personas	100d	58.0% dir.
TEPC full	80d	77.5% breakout
TEPC persona blend	80d	78.75% Breakout

Source: repo plots, run summaries, and research paper artifacts

Metrics differ by structured memory, and TEPC adds nonlinear regime signal on top.